Module 3 Final Challenge

Provide the best answer for each of the following questions.

Time Remaining 01:14:11

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though — spend the full time and attempt every one of them!

1

Review Later

Find the vertical distance between two parallel lines given by the following equations:

$$y(x) = 5x - 11$$
$$g(x) = 5x + 6$$

2

Review Later

There are 100 cycles, all of which are bicycles (which have 2 wheels) or tricycles (which have 3) wheels. The cycles have 237 wheels altogether. How many bicycles are there?

3

Review Later

Suppose that the symbol $x \mathbin{@} y$ means $\frac{1}{x} + \frac{1}{y}$. If $3 \mathbin{@} (2 \mathbin{@} x) = 1$, what is x?

4

Review Later

There are three 2-digit numbers that have the property that if you subtract 3 times its tens digit from it, the result is the original 2-digit number with the digits swapped.

Find the sum of these three 2-digit numbers.

5

Review Later

What is the sum of all even 2-digit numbers?

6

Review Later

A square's area plus its perimeter equals 285. What is the side length of the square?

7

Review Later

Starting from a point on the ground, a ball is thrown straight up to a height of 1 meter. It bounces up and down vertically, each time reaching a maximum height that is $\frac{2}{3}$ its previous maximum height. How many total meters will the ball travel, if it keeps bouncing forever?

8

Review Later

Simplify $\sqrt{20+10\sqrt{3}}$ and rewrite it in simplest radical form as $\sqrt{a}\left(b+\sqrt{c}\right)$ by filling whole numbers into these blanks:

√ ____ × (____ + √ ____)

9

Review Later

If Brian bikes to school at a speed of 15 miles per hour, he will get to school two minutes early. If he bikes to school at a speed of 12 miles per hour, he will get to school two minutes late. What speed should he bike at so that he gets to school on time?

Your answer should be an improper fraction in simplest terms without spaces; e.g., if the answer is 13/2, you would put 13 in the first blank and 2 in the second blank.

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10

Review Later

Find the value of the positive number:

$$\sqrt{30+\sqrt{30+\sqrt{30+\cdots}}}$$

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

The sum of two numbers is 16 and the sum of their cubes is 1456. What is the product of the two numbers?

12

Review Later

Two numbers have a difference of 4 with an arithmetic mean 7 larger than their geometric mean.

Find their sum.

Your answer should be a fraction in simplest terms, e.g. if the answer were 1/2, you would put 1 in the first blank, and 2 in the second blank.

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13

Review Later

What is the slope of the curve $y = x^3$ at the point $(2, 8)$?

14

Review Later

Two friends, R.U. Cumming and Ida Ngo, live 7 miles apart. R. U. starts to walk toward Ida's house, texting her along the way. After receiving R.U.'s text, Ida decides to walk towards his house, at $\frac{6}{5}$ his speed. They meet somewhere in between, after R.U. has spent 3 times as long walking as Ida. How many miles did R. U. walk?

Your answer should be the number only, without units.

15

Review Later

A triangle has corners $A(0,0)$, $B(12,0)$, and $C(0,6)$. If the midpoint of AB is connected to C and the midpoint of AC is connected to B, they intersect at a point with coordinates (x,y). What is x?

16

Review Later

The length of the straight line between $(4,6)$ and $(x+1, 4x-2)$ is $\sqrt{151x}$.

What is the sum of the possible values of x?

17

Review Later

A straight line passes through the points $(-10, \frac{1}{3})$ and $(-2, \frac{17}{3})$. What is the x-intercept of another line perpendicular to this first line which passes through the midpoint between $(-10, \frac{1}{3})$ and $(-2, \frac{17}{3})$?

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

Find the smallest positive integer n such that $n^2 - 20n + 64$ is a positive square.

19

Review Later

What is the value of b that makes

$$42ab + 24b = 35a + 20$$

true for all values of a?

Your answer should be a fraction in simplest terms; e.g. if the answer is $\frac{1}{2}$, you would put 1 in the first blank and 2 in the second blank.

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20

Review Later

Suppose $xy = 9$ and $x + \frac{1}{y} = 100$. Find the value of $y + \frac{1}{x}$.

Your answer should be a fraction in simplest terms, e.g. if the answer were 1/2, you would put 1 in the first blank, and 2 in the second blank.

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View Marked Questions

Finished

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